25900254 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025 Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBOT200,MG3DSCBBT200 MICROBIOLOGY AND PHYCOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type question

Answer any **10** questions

Each question carries **1** marks

1. What are pili? [U] / [CO1]
2. What is the function of bacterial flagella? [U] / [CO1]
3. What is the function of tail fibre in λ phage? [K] / [CO1]
4. Mention one microorganism that plays a significant role in environmental conservation. [K] / [CO1]
5. Define coenobium. [K] / [CO2]
6. State two major features of Lee's 2016 classification system for algae. [K] / [CO2]
7. Which structure in *Sargassum* helps it float? [K] / [CO2]
8. Name the haploid generation in the life cycle of algae. [U] / [CO2]

9. Which economically valuable product is extracted from *Gelidium* and *Gracilaria*? [U] / [CO3]

10. What is SCP? [K] / [CO3]

11. What is a 'photobioreactor' in algal cultivation? [K] / [CO3]

12. Name two commercial products obtained from macroalgae. [K] / [CO3]

[1x10 = 10]

Part B

Short Essay type question

Answer any **6** questions

Each question carries **5** marks

13. Discuss the advantages of biopesticides over chemical pesticides. [U] / [CO1]

14. Distinguish between generalized and specialized transduction. [U] / [CO1]

15. Write short notes on terrestrial and aquatic forms of algae. [K] / [CO2]

16. Describe the main features of the vegetative structure in *Chara.* [U] / [CO2]

17. Describe the main features of a *Cladophora* cell. [U] / [CO2]

18. Give a brief overview of the evolutionary sequence of algae as suggested by Fritsch. [U] / [CO2]

19. Discuss the significance of algal-fungal symbiosis in lichens. [U] / [CO3]

20. How do algae contribute to global primary production? [K] / [CO3]

[5x6 = 30]

Part C

Essay type question

Answer any **1** questions

Each question carries **10** marks

21. Compare and contrast viruses, viroids and prions. [U] / [CO1]

22. What are the advantages and disadvantages of water blooms? [K] / [CO3]

[10x1 = 10]